

Healthcare Encounter Performance Analytics

DESCRIPTIVE & DIAGNOSTIC ANALYTICS

Healthcare Encounter Performance Analytics

Power BI group report | 2019–2021 | multi-page dashboard and stakeholder story

1. Purpose, Scope and Method

This project converts 1,500 healthcare encounter records into a collaborative Power BI dashboard for descriptive and diagnostic analysis. The dashboard answers four stakeholder questions: who the patients are, which clinical categories show elevated risk, how facilities and providers differ, and what drives charges, cost and margin. The analysis covers 2019–2021 and includes 479 patients, 100 providers and 10 facilities.

1,500	15.27%	\$36.29M	39.61%
ENCOUNTERS	READMISSION RATE	TOTAL CHARGES	GROSS MARGIN

Data preparation and model

Power Query was used to standardize data types and categorical labels, review nulls and empty values, validate dimension keys, and create reporting attributes such as month labels and patient profiles. FactEncounter remains at one row per encounter. Seven dimension tables filter the fact table through single-direction one-to-many relationships, preserving a clear star schema and preventing ambiguous filter paths.

Model component	Relationship / mapping	Analytical role
Calendar Table	DateID → AdmissionDateID	Monthly/annual trends and date slicers
Patient Table	PatientID	Patients, gender, age group and state
Diagnosis + Procedure	DiagnosisID; ProcedureID	Clinical category and procedure drivers
Provider + Facility	ProviderID; FacilityID	Specialty, quality, capacity and drill-through
Payer Table	PayerID	Payer mix, charges, cost and margin
Medical Care CPI	YearMonthKey via TREATAS	Disconnected external benchmark for inflation

Measure and validation approach

Core DAX measures use distinct counts, sums, averages and DIVIDE for stable rates. Diagnostic measures remove and reapply procedure filters to compare Imaging with Non-Imaging, while a sufficient-volume measure suppresses groups below 20 encounters. The external Medical Care CPI table remains disconnected; TREATAS maps Calendar Table[YearMonthKey] to the CPI YearMonthKey inside the measure, so existing model transformations and relationships remain intact.

Validation totals: 1,500 encounters; 229 readmissions; \$36,294,228 charges; \$21,917,598 cost; \$14,376,630 margin.

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2. Dashboard Design and User Experience

The four analytical pages use a consistent question-led layout: page-specific slicers on the left, headline measures across the top, and explanatory visuals below. The pages intentionally use different slicers and are not globally synced; each page therefore preserves the context needed for its own stakeholder question.

Patient Overview

Clinical / Diagnosis Insights

Demographics, geography and activity

Procedure and readmission drivers

Provider and Facility Performance

Financial / Payer Analysis

Quality, capacity and specialty metrics

Margin bridge, payer mix and CPI benchmark

Visual selection and insight

Visual type	Why it was chosen	Insight conveyed
KPI cards	Fast status check	Patients, encounters, readmissions, LOS, charges, cost and margin.
Combo + indexed lines	Trend and benchmark	Shows monthly activity/readmission movement and compares average charge with Medical CPI (2019 = 100).
Bar + stacked column	Rank and composition	Compares clinical rates, demographics, facilities, payer economics and bed capacity.
Map	Spatial pattern	Locates patient concentration across states.
Scatter + matrix	Relationship and detail	Tests LOS–cost association; bookmark switches to diagnosis-by-procedure detail.
Tree + waterfall	Diagnosis and reconciliation	Decomposes readmission drivers and reconciles charges minus cost to margin.

Interactive UX. Two clinical bookmarks switch between the LOS–cost relationship view and diagnosis-by-procedure detail without resetting slicers (Data is off). A right-click on a facility bar opens the hidden Facility Details page and passes the selected FacilityID and source context; the Back button returns to the source page.

3. Diagnostic Analytics: Evidence and Possible Causes

The investigation used DAX measures, filter isolation, a minimum-volume rule, the decomposition tree and facility drill-through. Findings are associations within this dataset; they identify where stakeholders should investigate, not causal effects.

Procedure → payer diagnostic path

Facility 6 drill-through: 166 encounters, 18.07% readmission

Finding 1 — Imaging encounters have elevated readmission

Cohort	Encounters	Readmissions	Rate
Non-Imaging	1,181	165	13.97%
All clinical	1,500	229	15.27%
Imaging	319	64	20.06%
Imaging + Private payer	58	15	25.86%

Imaging exceeds Non-Imaging by 6.09 percentage points (about 44% relatively). The tree identifies Private as the highest Imaging payer subgroup. Within Imaging + Private, Acute is 26.67% (8/30) and Chronic is 25.00% (7/28), so the elevated rate persists across both broad diagnosis categories rather than being explained by diagnosis mix alone.

Finding 2 — Facility 6 materially exceeds Facility 4

Facility	Encounters	Readmissions	Rate	Avg LOS
Facility 4	155	17	10.97%	7.01
All facilities	1,500	229	15.27%	7.62
Facility 6	166	30	18.07%	7.67

The 7.10-point gap is visible despite similar encounter volumes (166 versus 155), making a very small denominator an unlikely sole explanation. Facility 6 drill-through shows 77 Lab, 55 Surgery and 34 Imaging encounters and an average charge of \$24.03K; the next diagnostic step is case-mix and provider-level review rather than assuming poor performance.

Contextual trend — average charges lagged Medical CPI

Year	Avg charge / encounter	Charge index	Medical CPI index
2019	\$23,904.55	100.00	100.00
2020	\$24,664.52	103.18	104.11
2021	\$23,969.00	100.27	105.39

From 2020 to 2021, nominal average charge declined 2.82% while Medical CPI rose 1.23%, implying an approximately 4.0% decline relative to the CPI benchmark. This supports monitoring real pricing pressure alongside nominal revenue.

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4. Recommendations, Limitations and Conclusion

The dashboard is most useful as a prioritization tool: it identifies where a clinical, operational or financial review is likely to yield the greatest value.

Recommendations

Recommendation 1 — Review the Imaging care pathway.

Start with the 64 Imaging readmissions and the Private payer subgroup (25.86%). Review discharge instructions, follow-up scheduling, repeat imaging, diagnosis detail and whether the readmission occurred within the organization's standard window.

Recommendation 2 — Investigate Facility 6 and benchmark Facility 4.

Use the drill-through page to compare procedure, diagnosis, payer and provider mix. Replicate relevant practices from Facility 4 only after risk adjustment confirms that case severity and patient mix do not explain the gap.

Recommendation 3 — Manage charges in real terms.

Track the average-charge index against Medical CPI each month and by payer. The 2021 divergence suggests reviewing payer contracts, charge schedules and cost containment before interpreting nominal totals as improved performance.

Recommendation 4 — Embed a governed monthly review.

Retain the 20-encounter threshold, validate refresh totals, record selected slicers, and use bookmarks and drill-through as a repeatable investigation workflow. Add severity, comorbidity and reimbursement fields when available.

Limitations

The dataset contains only 1,500 encounters from 2019–2021 across 10 facilities, so subgroup estimates may still be unstable. The model lacks illness severity, comorbidities, staffing, detailed diagnosis/procedure codes, actual payer reimbursement and a defined readmission interval. Facility and clinical comparisons are therefore unadjusted and should not be interpreted causally. Medical Care CPI is a national consumer benchmark rather than an organization-specific hospital input-cost or reimbursement index; TREATAS provides monthly alignment but does not establish a business relationship.

Requirement completion

Requirement	Implementation	Status
Dashboard	4 required pages + home page	Met
Visual design	Cards, lines, bars, map, scatter, matrix, tree, waterfall	Met

UX	Page slicers, two clinical bookmarks, facility drill-through	Met
Diagnostics	Two investigated anomalies plus CPI contextual trend	Met
External data	BLS Medical Care CPI-U, cited below	Met

Conclusion

The completed report meets the required dashboard, advanced visual, UX and diagnostic components. Its strongest evidence is the elevated Imaging readmission pattern and the Facility 6 performance gap, while the CPI comparison adds an external economic context for financial interpretation.

External data source: U.S. Bureau of Labor Statistics. (n.d.). Consumer Price Index for All Urban Consumers: Medical care in U.S. city average (Series CUUR0000SAM) [Data set]. Monthly observations used for 2019–2021. [BLS CPI databases](#) | [Series CUUR0000SAM](#)